



PRO 2

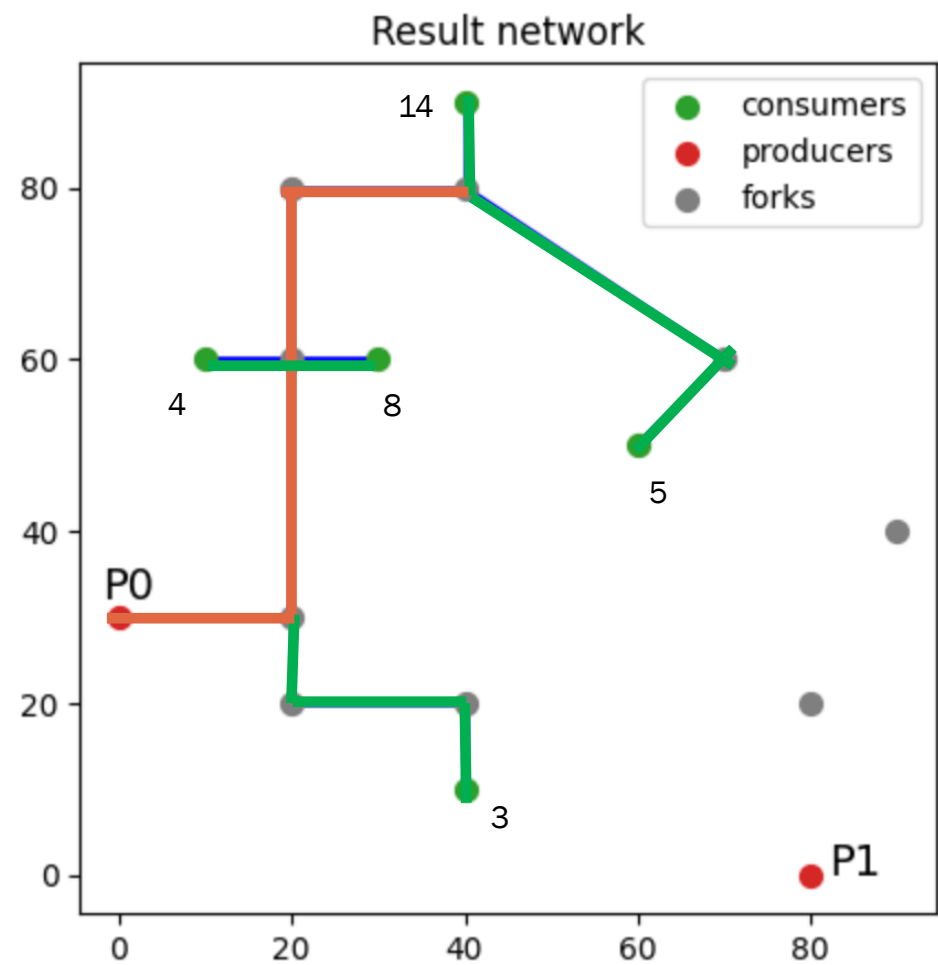
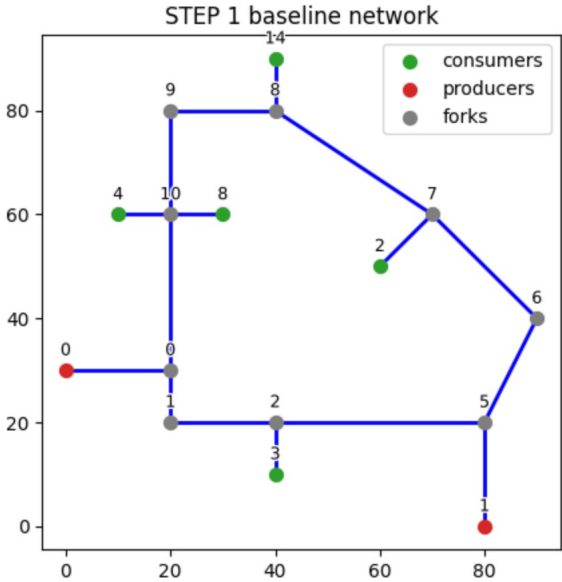
MJ2505 District Heating Network Optimization

Group 14

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Step 1: Base DH Network

Load number	Peaks (kW)
Load 3	74.00
Load 2	50.50
Load 14	150.50
Load 4	108.93
Load 8	100.50



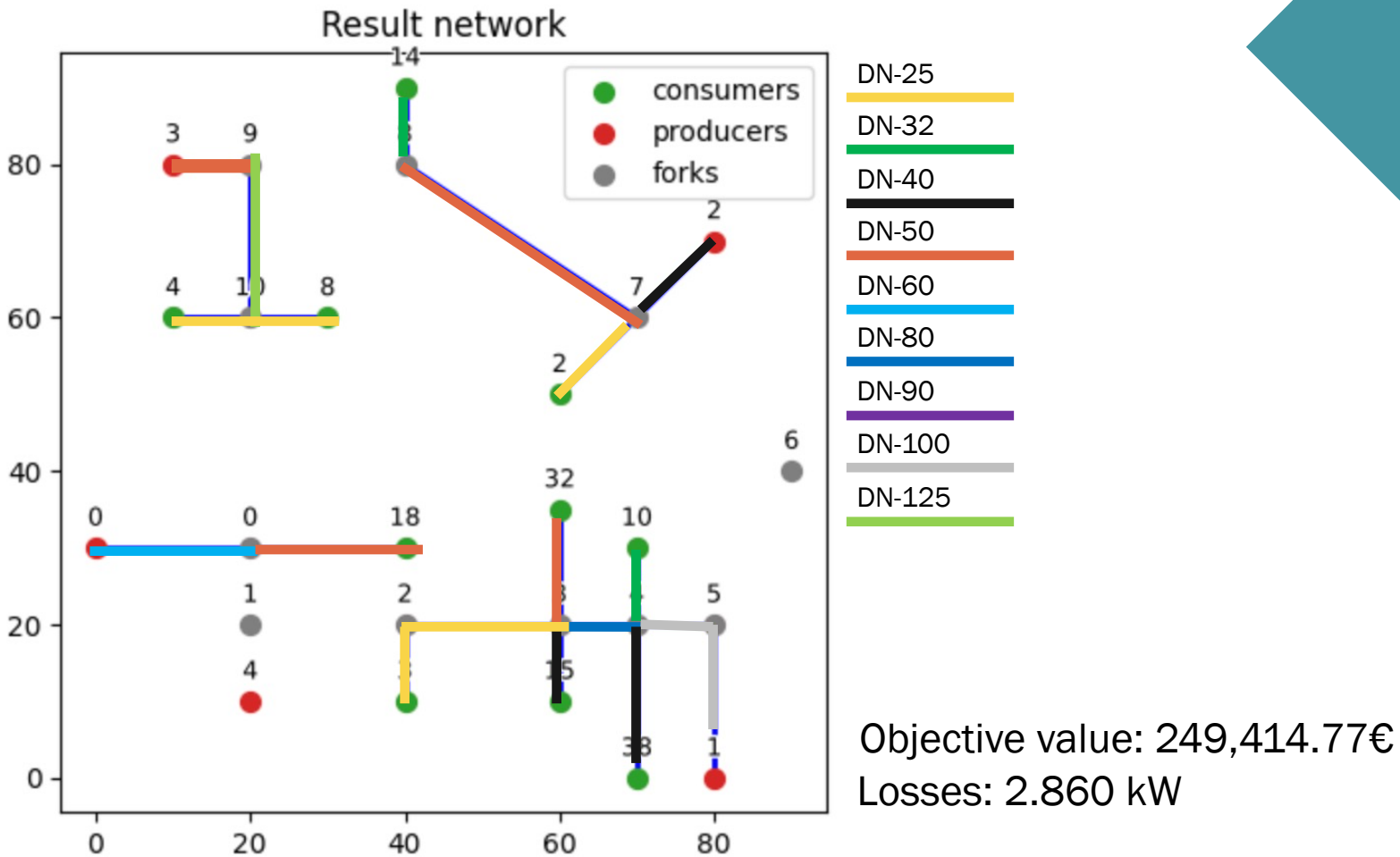
DN-32
DN-50

Objective value: 117455.01€
Losses: 1.852 kW

label_3	nonconvex	l_factor	l_factor_fix	cap_max	cap_min	capex_pipes	fix_costs
DN-32	1	0	0.0086	200	53	0.5	491
DN-50	1	0	0.0091	500	178	0.5	563

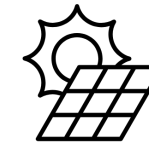
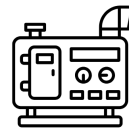
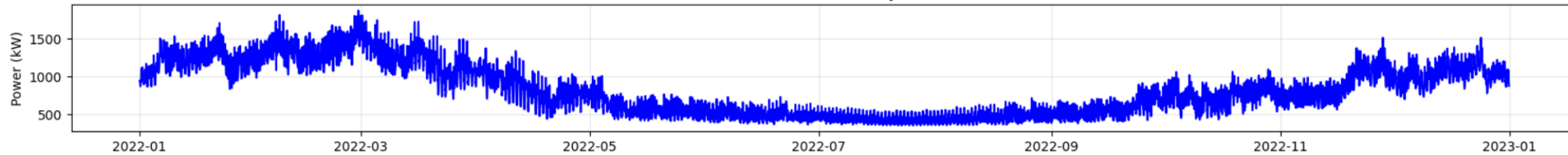
Step 2: Expanded DH network

Load number	Peaks (kW)
Load 3	74.00
Load 2	50.50
Load 14	150.50
Load 4	108.93
Load 8	100.50
Load 15	317.00
Load 32	491.00
Load 38	258.00
Load 10	158.93
Load 18	358.00



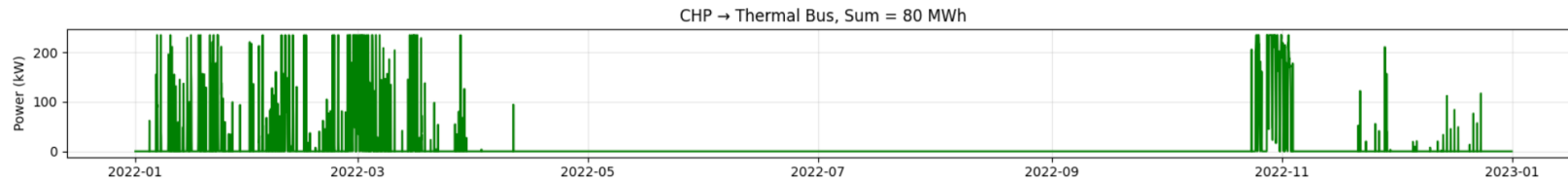
Step 3: Demand & Configuration

Peak Demand [kW(th)]	Total Demand (yearly) [MWh(th)]
1869.59	6865.5

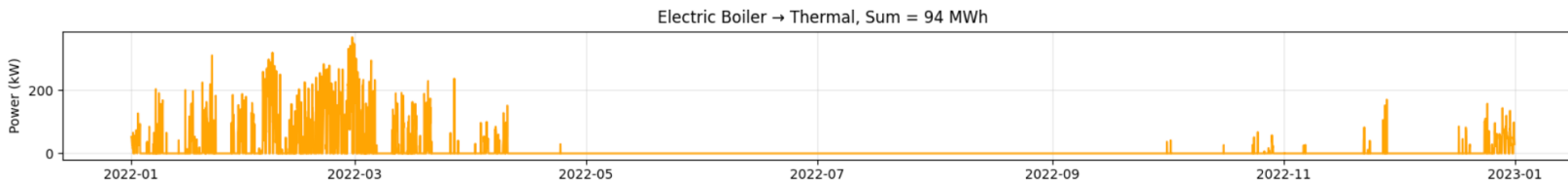


Technology	NG CHP	Heat Pump	Electric Boiler	Solar Collectors	Biomass CHP
Specific CAPEX [€/kW]	1200	1400	250	300	2200
O&M costs [€/kWh]	0.018	0.002 (1.5% CAPEX)	0.0003 (1% CAPEX)	0.0005 (1.5% CAPEX)	0.009 (3% CAPEX)
Capacity limits [kWth] (min – max)	100 - 800	100 – 500	100 – 800	802.09 (Fixed)	100 – 800
Final Capacity [kWth]	234.55	500	366.97	802.09	717.46

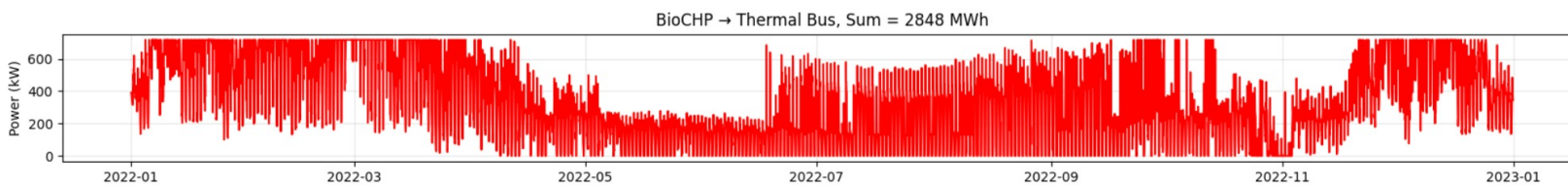
Step 3: Visual Results - thermal



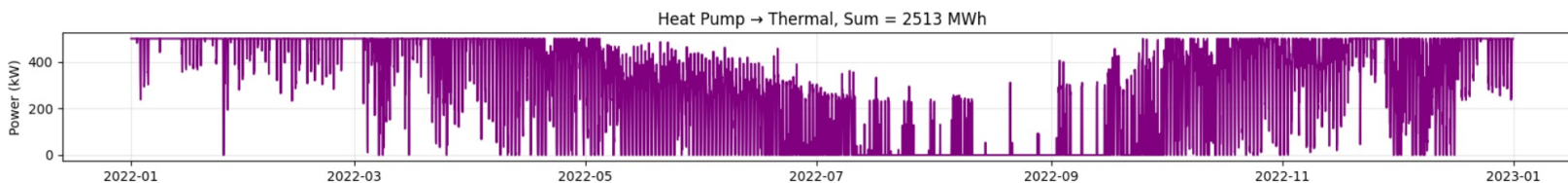
117 hours at cap
longest run 16h



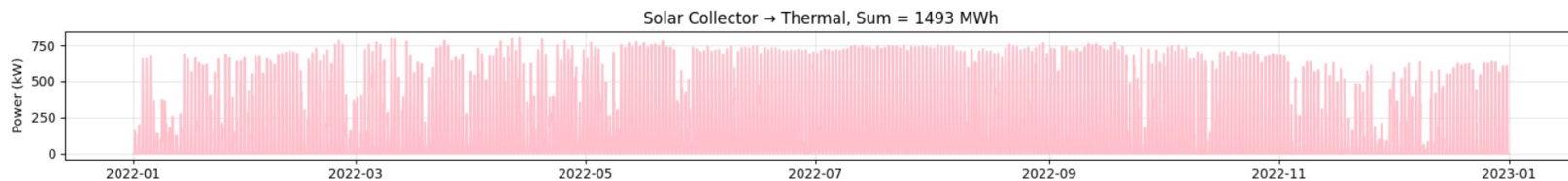
1 hours at cap
longest run 1h



1066 hours at cap
longest run 41h

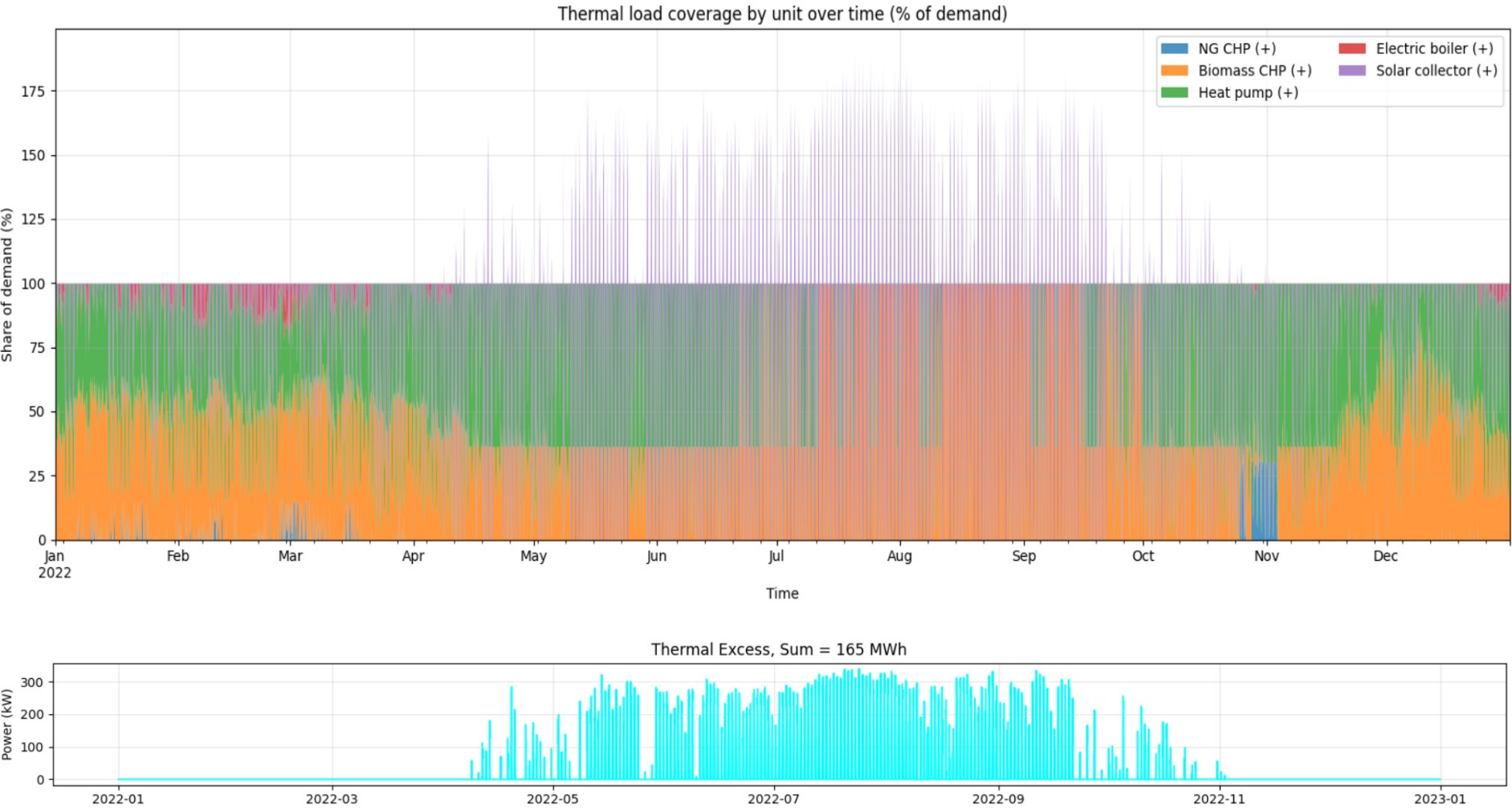


2825 hours at cap
longest run 139h



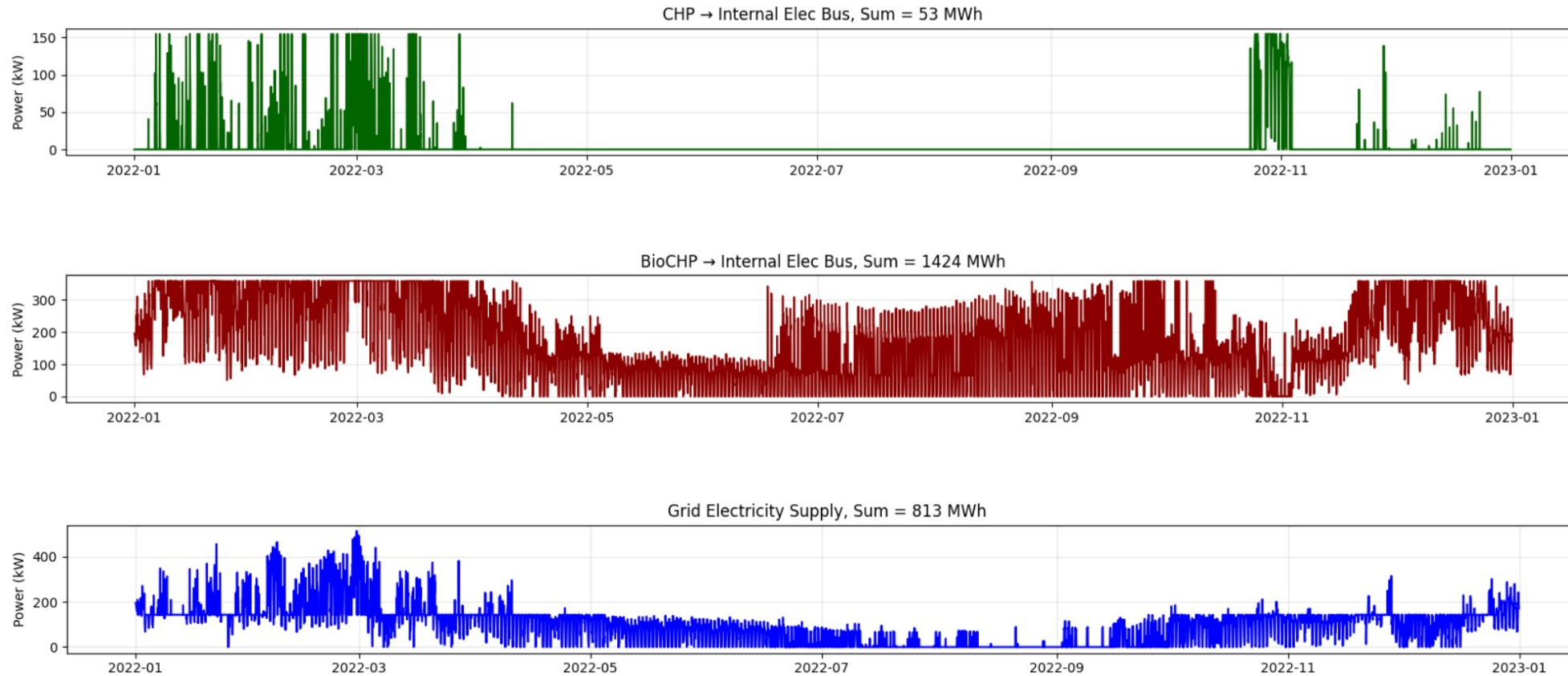
1 hours at cap
longest run 1h

Step 3: Visual Results - thermal



Technology	Demand share (%)
Biomass CHP	41.5
Gas CHP	1.2
Heat Pump	36.6
Electric Boiler	1.5
Solar Collector	19.2

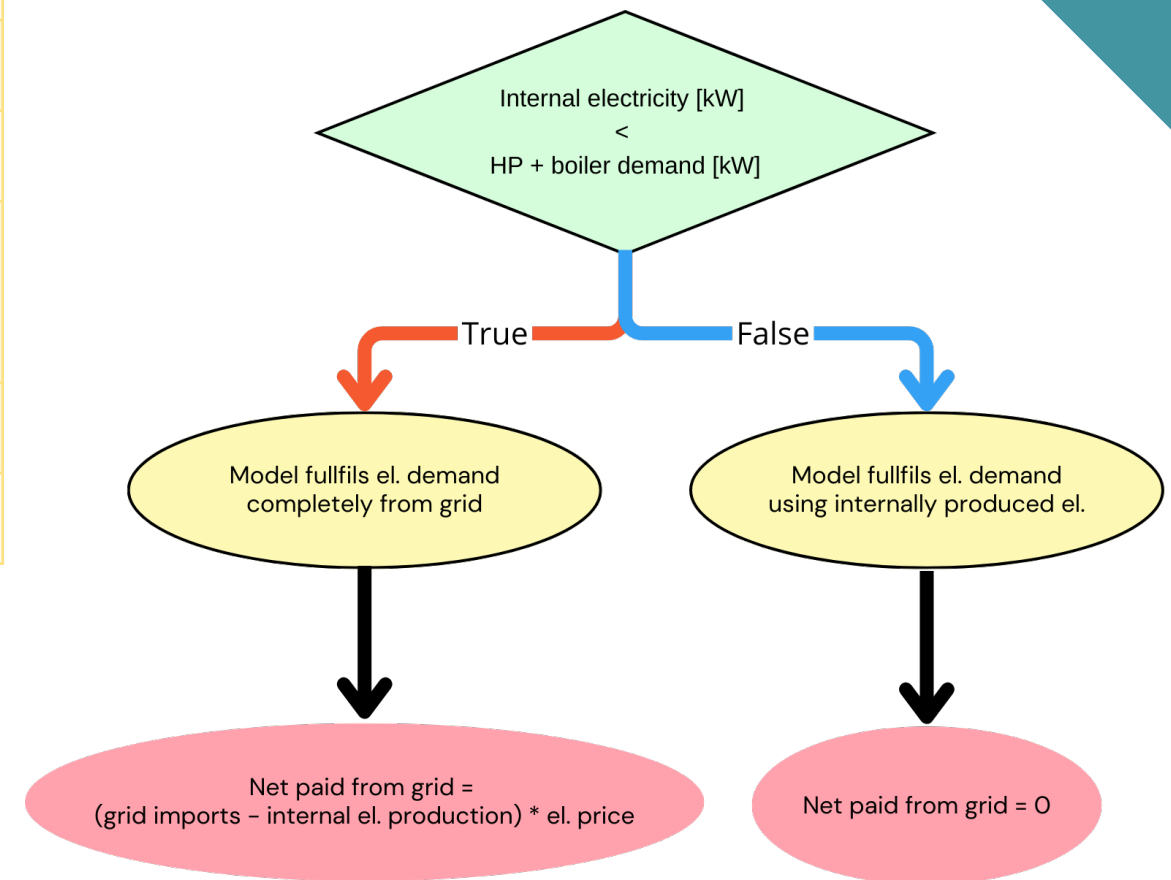
Step 3: Visual Results - electricity



Step 3: Economic Results

Technology	CAPEX [€]	O&M [€/yr]	Fuel [€/yr]
Biomass CHP	1,578,412	25,000	246,123
Gas CHP	281,454	1,433	12,859
Heat Pump	700,000	6,026	84,863 (electricity)
Electric Boiler	91,741	27	
Solar Collector	240,626	768	NA
Totals	2,892,233	377,099	

How are grid imports [€] calculated?



Step 4: Energy System

Technology	Capacity [kWth]	Δ Capacity	Production [MWh]	Δ Production
Biomass CHP	721	+2.9 (+0.4%)	2866.1	17.22 (+0.60%)
Gas CHP	233	- 0.6 (- 0.2%)	81.0	0.08 (+0.10%)
Heat Pump	500	0 (0%)	2520.6	6.88 (+0.27%)
Electric Boiler	368	+1 (+0.2%)	94.8	0.78 (+0.83%)
Solar Collector	802.1	0 (0%)	1493.8	0 (0%)



Increased Demand due to Losses

From 6865.5 to 6893.6 MWh
(+28.2/+0.4%)



Increased Heat Generation

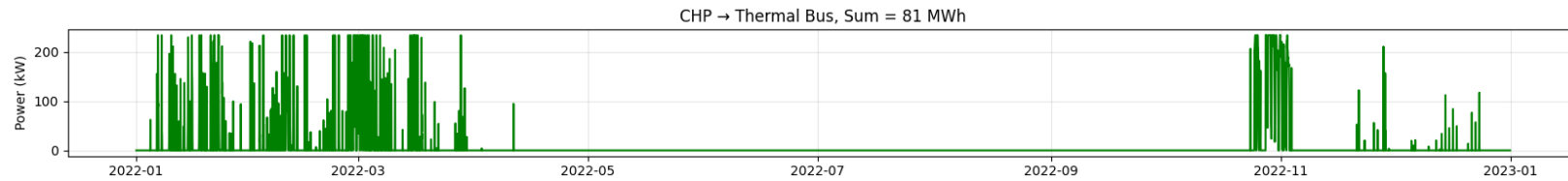
From 7031.3 to 7056.3 MWh
(+25.0/+0.4%)



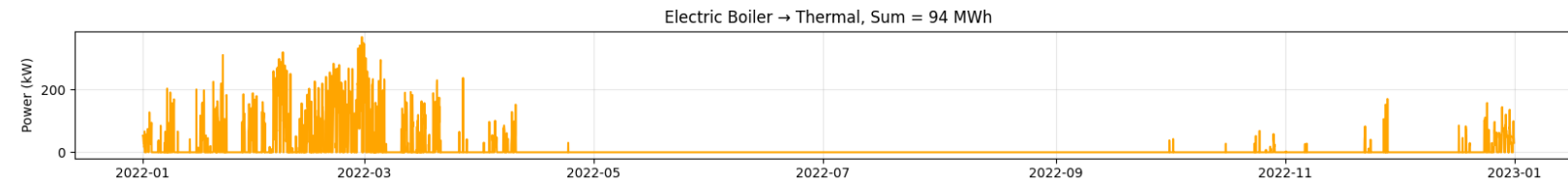
Thermal Surplus reduced by 3 MWh

System efficiency increased 0.1%
(97.7%)

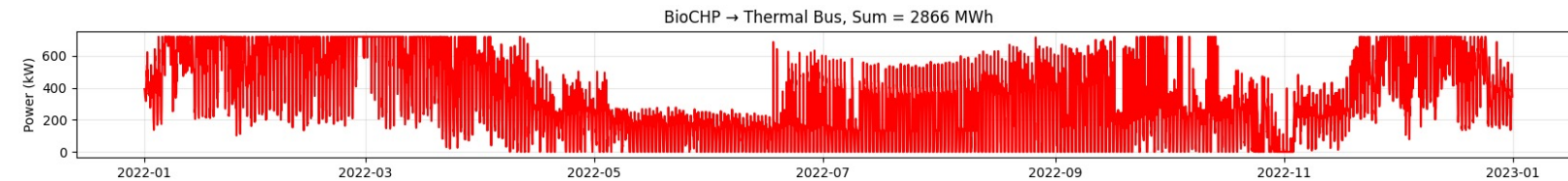
Step 4: Visual Results - thermal



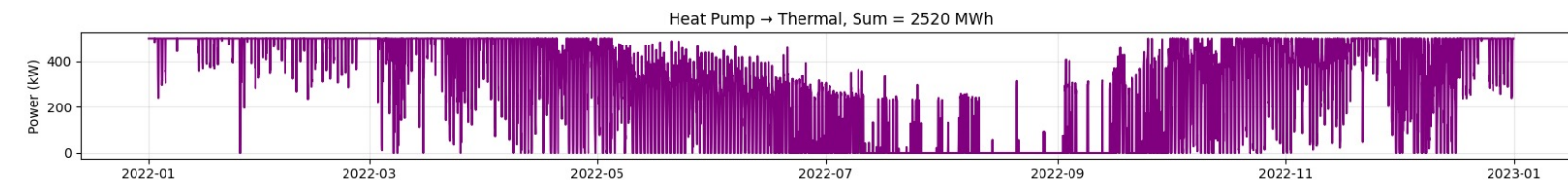
122 hours at cap
5 hours more



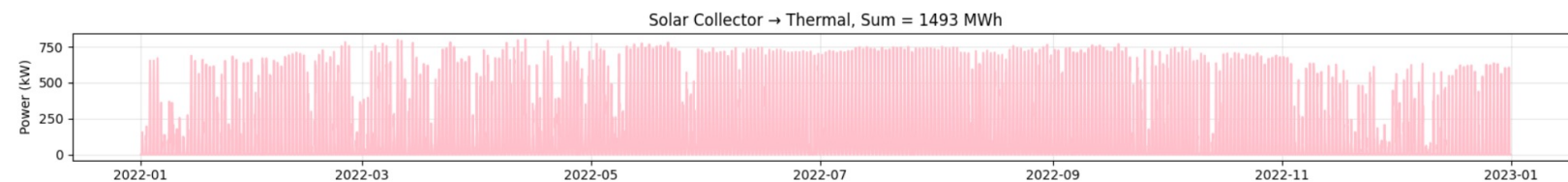
1 hours at cap
Same as Step 3



1068 hours at cap
2 hours more



2859 hours at cap
34 hours more



1 hours at cap
Same as Step 3

Step 4: Economic Results

Technology	CAPEX [€]	Δ CAPEX Step 3 [€]	OPEX (O&M + Fuel) [€/yr]		Δ OPEX Step 3 [€]	
Biomass CHP	1,584,605	+5,357	272,762		+1,639	
Gas CHP	280,734	- 309	14,302		+10	
Heat Pump	700,000	0	6,042	84,876 (electricity)	+16	+13 (electricity)
Electric Boiler	91,992	+259	27		0	
Solar Collector	240,626	0	768		0	
Totals	2,897,957	+5,308 (+0.18%)	378,777		+1,678 (+0.4%)	

Step 5: Scenario A

Technology	Biomass CHP	Gas CHP	Heat Pump	Electric Boiler	Solar Collectors
Capacity [kWth]	721	233	500	368	802.1

Scenario A: fuel costs + CO₂ emission cost

Grid emission factor = 0.35 kgCO₂/kWh

Natural gas emission factor = 0.2 kgCO₂/kWh

Biomass emission factor = 0

Unit	Step 4 [MWh]	Scenario A [MWh]	Δ Production [MWh]	Δ [%]	Demand share [%]
NG CHP	81	49	- 32	- 39.5	0.7
Electric boiler	95	79	- 16	- 16.8	1.2
Heat pump	2521	2439	- 82	- 3.3	35.4
Solar collector	1493	1493	0.00	0.00	19.3
Biomass CHP	2867	2993	+126	+4.39	43.4

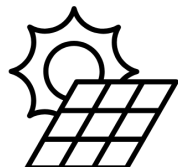
Step 5: Scenario B

Biomass CHP



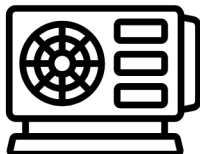
63.9%

Solar Collectors



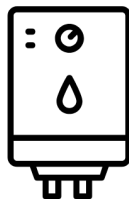
19.3 %

Heat Pump



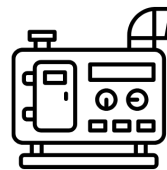
15.2 %

Electric Boiler



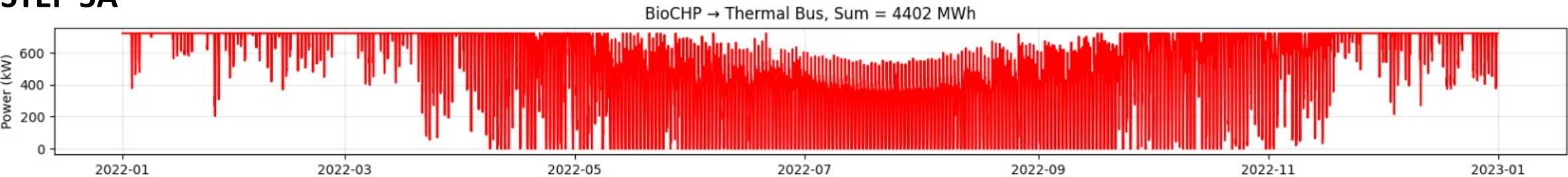
1.5 %

NG CHP

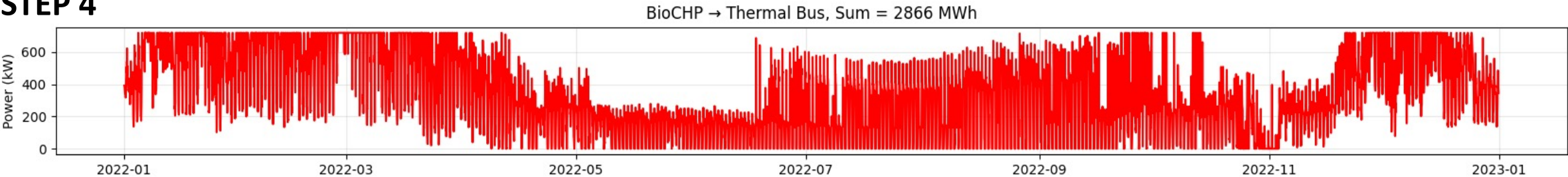


0.1 %

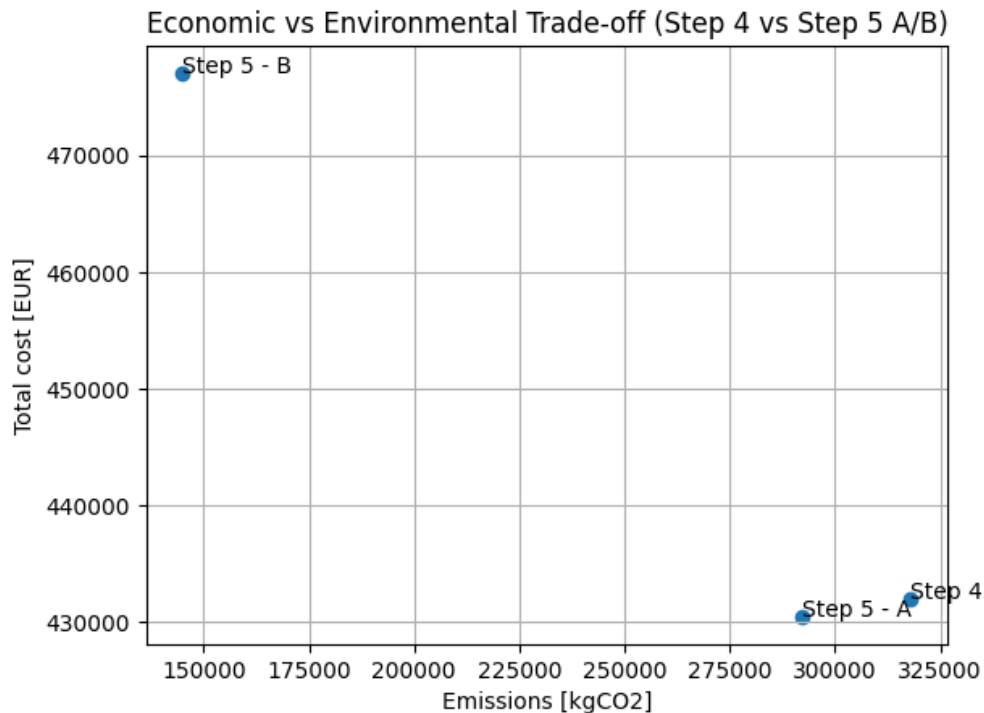
STEP 5A



STEP 4



Step 5: Economic vs Environmental dispatch



Scenario	Total Cost [k€]	Emissions [tCO ₂]	Δ Cost vs Step 4 [k€]	Δ Emissions vs Step 4 [tCO ₂]
Step 4	431.9	318.0	0.0	0.0
Step 5A	430.4	292.0	-1.5	-25.9
Step 5B	477.0	145.0	+45.1	-173.0

Total Cost = Electricity Cost + Biomass Cost
+ Gas Cost + Carbon Cost

Discussion and improvements

- Solar Collectors have to be manually implemented in the utilized .oemof version
- In reality, dispatchable units such as CHPs operate within a limited power range, introducing minimum load constraints would prevent unrealistically low part-load operation.
- Converters can't draw energy simultaneously from multiple buses. To avoid unrealistic full reliance on the grid when internal supply is partially available, a buffer system should be implemented



Thank you

Group 14

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References

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